

ANDREAS PANAYI

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EDUCATION

National and Kapodistrian University of Athens (NKUA)	2024 - 2026
National Technical University of Athens (NTUA)	
M.S. in Theoretical Computer Science (ALMA)	GPA: 9.89/10
Track: Algorithms, Logic, and Discrete Mathematics	
National and Kapodistrian University of Athens (NKUA)	2020 - 2024
B.Sc. in Mathematics	GPA: 8.84/10

RESEARCH INTERESTS

Theoretical Computer Science; Algorithms (fine-grained, parameterized); Advanced Data Structures (dynamic, geometric); Graph Theory (extremal, structural); Computational Complexity, Database Theory

RESEARCH EXPERIENCE

Research and Innovation Foundation of Cyprus	Jun 2026 - Jul 2026
<i>Research Intern</i>	

- Conducted industry research bridging Big Data and Machine Learning, developing algorithmic and machine learning methods to drive a novel end-to-end database compression system.
- Led the design and optimization of the underlying system, from theoretical formulation to proof-of-concept implementation with running time analysis and experimentation.

Archimedes Research Unit, Athena Research Center	Jun 2025 - Jan 2026
<i>Intern Researcher</i>	

- Designed and implemented advanced data structures for computational geometry problems, focusing on skyline (Pareto-front/maxima) reporting.
- Authored an extended survey on models of computation and advanced data structures, spanning computational-geometry trees, external-memory structures, and van Emde Boas-based techniques, culminating in an implementation and analysis of the Micro-Macro technique.

TEACHING EXPERIENCE

National and Kapodistrian University of Athens	2021 - 2026
<i>Teaching Assistant, Computer Science I</i>	

- Conducted undergraduate laboratory sessions for the mandatory course *Computer Science I* in the Department of Mathematics.
- Taught programming in MATLAB (2021 - 2024) and Python (2024 - 2026), as well as $\text{\LaTeX}$, computer arithmetic, and numerical linear algebra.

RESEARCH PAPERS

- **Listing Even Cycles Faster than the Submodular-Width Barrier.**
Vasileios Nakos, Hung Q. Ngo, and Andreas Panayi. *arXiv preprint* arXiv:2605.30564, 2026.
- **Faster Even-Cycle Listing via Unbalanced Supersaturation.**
Andreas Panayi, National and Kapodistrian University of Athens, 2026. Pergamos: uoadl:5402752.

SELECTED COURSEWORK

Advanced Algorithms; Advanced Data Structures; Graph Theory; Advanced Computational Complexity; Theory of Computation; Machine Learning; Statistical Learning; Algorithmic Data Science

TECHNICAL SKILLS

Programming Languages: Python, Go, TypeScript, C++, MATLAB, Julia, R, Haskell, Prolog

Tools: Git, GitHub, Docker, SSH, Linux, L^AT_EX, X_YL^AT_EX

Languages: Greek (native), English (fluent)

AWARDS AND SCHOLARSHIPS

Cyprus State Scholarships Foundation 2020 - 2026

Full scholarship for the entire duration of bachelor's and master's studies

Hellenic Bank Cyprus 2020 - 2024

Full scholarship for undergraduate studies

Athletics Distinctions

Represented the Cyprus National Athletics Team in international competitions across all age categories, earning multiple distinctions in individual and relay events. Pancyprian champion and gold medalist in 100m, 300m hurdles, and 400m hurdles in school and club competitions.

ADDITIONAL INFORMATION

Military Service Jul 2019 - Sep 2020

Served in the technical department during 14 months of military service in Cyprus